

TRIAL: Token Relations and Importance Aware Late-interaction for Accurate Text Retrieval

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Text Retrieval

Introduction

Text retrieval/search serves as a **core component that enables efficient and reliable access** to digital information in our daily lives.

Many applications:

-  Web Search Engines
-  E-commerce Product Search
-  Legal Document Discovery
-  Medical Information Retrieval

Problem Definition

Introduction

Given a natural language question query Q and a text corpus $C = \{D_1, D_2, \dots, D_n\}$, the goal of text retrieval is to rank documents $D \in C$ by their *relevance* to query Q .

Text retrieval model is a **scoring function**:

$$\text{rank}(C \mid Q) = \underset{D \in C}{\text{argsort}} \text{score}(Q, D)$$

The relevance distribution over the corpus is **approximated via a softmax transformation** of the scores:

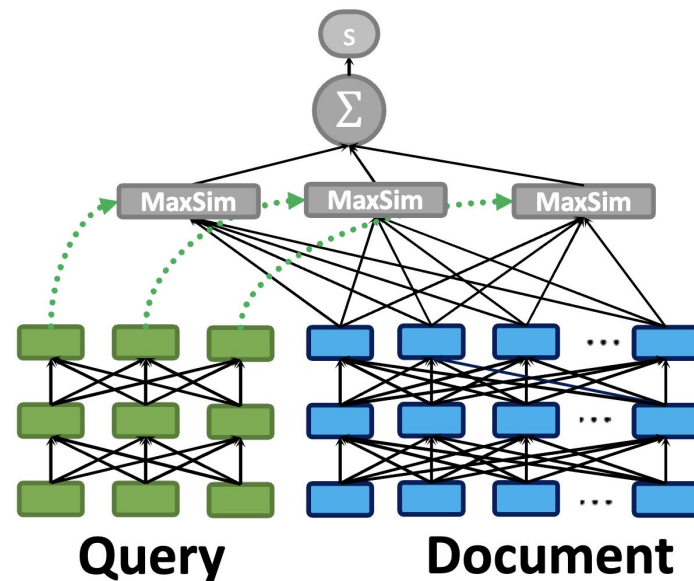
$$P(D \mid Q) = \underset{D \in C}{\text{softmax}}(\text{score}(Q, D))$$

Late-interaction

Background

- Late-interaction computes **similarity between bags of contextualized query and document embeddings**
- The score is obtained by **summing the maximum pairwise similarities** between each query token and all document token embeddings

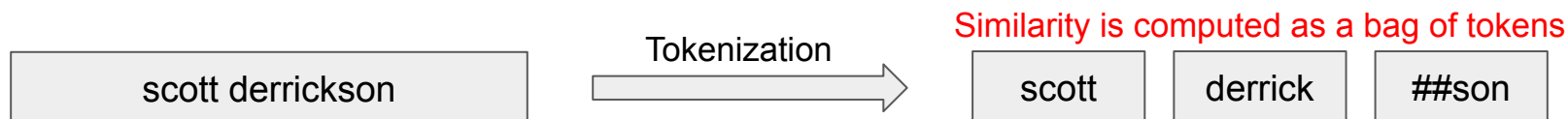
$$\text{score}(Q, D) = \sum_{i=1}^{|Q|} \max_{j=1}^{|D|} (E_{q_i} \cdot E_{d_j})$$



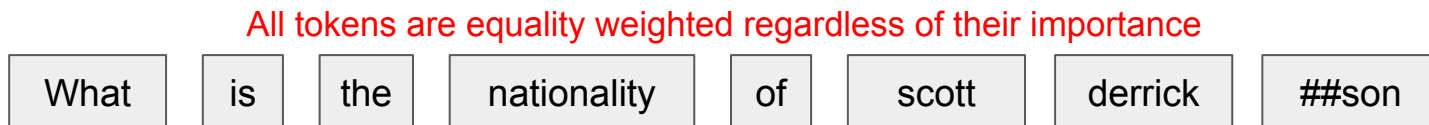
Limitations of Late-interaction

Background

- The naive token-level summation can lead to **incorrect relevance scores** when words or phrases are **fragmented into multiple tokens**.



- Equally weighted token-level summation **fails to account for term importance** (a factor emphasized in traditional methods such as TF-IDF and BM25).



TRIAL: Token Relations and Importance Aware Late-interaction

Method

- We propose an advanced scoring function for late-interaction that models **token relation score** and **token importance weight**.

$$\text{score}(Q, D) := \sum_{i=1}^{|Q|} \underbrace{w_{q_i}}_{\text{weighted sum based on query token importance}} \max_{j=1}^{|D|} \left(\text{sim}(E_{q_i}, E_{d_j}) + \lambda \underbrace{\text{sim}(\text{Rel}(q_i, q_{i-1}), \text{Rel}(d_j, d_{j_{i-1}}))}_{\text{consider relation between tokens (query-side bigrams)}} \right)$$

Token Relation Score

Method

- Token relation score **explicitly models the similarity between the token relations** in queries and documents.
- **Dot product** as the similarity function

$$\text{sim}(R_1, R_2) := R_1 \cdot R_2, \quad R_1, R_2 \in \mathbb{R}^d$$

- Encode the relation between two tokens by **concatenating** their embedding vectors and **passing through a two-layer Multi-layer Perceptron (MLP)**

$$\text{Rel}(t_1, t_2) := \text{LL}\left(\text{Mish}\left(\text{LN}\left(\text{LL}(E_{t_1}; E_{t_2})\right)\right)\right)$$

Token Importance Weight

Method

- The token importance weight **explicitly captures the varying significance of tokens** within a query.
- We employ a **two-layer gating neural network** that leverages Mish and ReLU activations to estimate token-wise importance:

$$\text{Gate}(t_i) := \text{ReLU} \left(\text{Mish} \left(W_1 E_{t_i}^\top + b_1 \right) W_2 + b_2 \right)$$

Training Retrieval Model with TRIAL

Method

- We train the retrieval model using **contrastive learning** with a **regularization loss**:

$$L_{\text{total}} = L_{\text{KL}} + L_{\text{CE}} + L_{\text{R}}$$

- The **KL divergence** and **cross-entropy losses** jointly help the model distinguish relevant documents from irrelevant ones:

$$L_{\text{KL}} := \sum_{D \in C} P_{\text{teacher}}(D | Q) \log \left(\frac{P_{\text{teacher}}(D | Q)}{P_{\text{student}}(D | Q)} \right) \quad L_{\text{CE}} := -\log \left(\frac{e^{\text{score}(Q, D^+)}}{e^{\text{score}(Q, D^+)} + \sum_{D^- \in N} e^{\text{score}(Q, D^-)}} \right)$$

- We introduce a **regularization loss** that optimizes the gating function to emphasize important tokens:

$$L_{\text{R}} := \lambda_q \|W_Q\|_1$$

Experimental Setting

Experiments

- **Evaluation benchmarks**

- In-domain: MSMARCO
- Out-of-domain: BEIR and LoTTE

- **Evaluation metrics**

- nDCG@10 for MSMARCO and BEIR
- Success@5 for LoTTE

- **Evaluation environment**

- Ubuntu server (Intel Xeon Gold 6230 CPU)
- 1TB RAM, 1TB storage, and four NVIDIA RTX A6000 GPUs

Retrieval Accuracy

Experiments

- TRIAL shows **state-of-the-art retrieval accuracy** both in MSMARCO and BEIR benchmark datasets.

	MS MARCO	BEIR													Avg.
		Arguana	Climate-fever	DBpedia-entity	Fever	Fiqa	HotpotQA	NFCorpus	NQ	Quora	SciDocs	SciFact	TREC-COVID	Touche2022	
Sparse Retrieval															
BM25	22.8	31.5	21.3	31.3	75.3	23.6	60.3	32.5	32.9	78.9	15.8	66.5	65.6	36.7	44.02
SPLADEv2	43.3	47.9	23.5	43.5	78.6	33.6	68.4	33.4	52.1	83.8	15.8	69.3	71.0	27.2	49.85
SPLADE++	-	-	-	-	-	-	-	-	-	-	-	-	-	-	50.70
Full-interaction based Dense Retrieval															
MonoBERT	-	29.1	11.2	43.6	74.8	38.9	66.0	36.2	52.3	77.3	15.2	66.9	72.7	31.7	47.38
CELI	-	44.3	15.2	44.9	74.0	40.1	71.4	36.8	52.7	69	16.2	71.5	73.6	32.0	49.36
Representation based Dense Retrieval															
DPR	17.7	17.5	14.8	26.3	56.2	11.2	39.1	18.9	47.4	24.8	7.7	31.8	33.2	13.1	26.31
ANCE	38.8	41.5	19.8	28.1	66.9	29.5	45.6	23.7	44.6	85.2	12.2	50.7	65.4	24.0	41.32
GTR	42.0	51.1	24.1	34.7	66.0	34.9	53.5	30.8	49.5	88.1	14.9	60.0	53.9	20.5	44.77
Late-interaction based Dense Retrieval															
ColBERT	40.1	23.3	18.4	39.2	77.1	31.7	59.3	30.5	52.4	85.4	14.5	67.1	67.7	20.2	45.14
ColBERTv2	-	46.3	17.6	44.6	78.5	35.6	66.7	33.8	56.2	85.2	15.4	69.3	73.8	26.3	49.95
XTR	45.0	40.7	20.7	40.9	73.7	34.7	64.7	34.0	53.0	86.1	14.5	71.0	73.6	34.7	49.41
LITE	45.2	42.4	21.3	43.4	78.8	33.6	68.1	35.8	54	83.9	16.4	63.3	76.3	30.5	49.83
TRIAL	46.3	53.9	18.2	50.7	78.2	34.6	66.9	33.3	57.2	82.4	14.8	65.7	77.3	31.0	51.09
TRIAL w/o relations	46.1	50.4	18.5	49.8	77.2	34.0	63.3	32.2	57.0	80.9	13.8	63.9	77.3	29.8	49.85
TRIAL w/o rel. & importance	45.6	49.6	18.0	48.8	75.9	32.0	62.2	32.3	51.9	80.6	14.0	63.3	77.3	29.6	48.88

Retrieval Accuracy (cont'd)

Experiments

- TRIAL shows **state-of-the-art retrieval accuracy** in LoTTE search dataset, while showing comparable accuracy in LoTTE forum dataset.

Table 2: Success@5 on the LoTTE search dataset

	Writing	Recreation	Science	Technology	Lifestyle	Pooled	Avg.
BM25	60.3	56.5	32.7	41.8	63.8	48.3	50.56
ColBERT	74.7	68.5	53.6	61.9	80.2	67.3	67.70
ColBERTv2	80.1	72.3	56.7	66.1	84.7	71.6	71.91
XTR	77.0	69.4	54.9	63.2	82.1	69.0	69.26
SPLADEv2	77.1	69.0	55.4	62.4	82.3	68.9	69.18
TRIAL	79.5	73.2	57.6	67.2	83.6	71.8	72.15

Table 3: Success@5 on the LoTTE forum dataset

	Writing	Recreation	Science	Technology	Lifestyle	Pooled	Avg.
BM25	64.0	55.4	37.1	39.4	60.6	47.2	50.61
ColBERT	71.0	65.6	41.8	48.5	73.0	58.2	59.68
ColBERTv2	76.3	70.8	46.1	53.6	76.9	63.4	64.51
XTR	73.9	68.7	42.2	51.9	74.4	60.1	61.86
SPLADEv2	73.0	67.1	43.7	50.8	74.0	60.1	61.45
TRIAL	75.5	69.8	45.3	54.0	75.8	62.3	63.78

Retrieval Speed

Experiments

- Despite the additional computations for token relations and importance, TRIAL achieves **comparable retrieval speed** to ColBERTv2.

Table 5: Execution time (ms) for each stage of the retrieval process. The time is measured over 1,000 sampled queries from MSMARCO.

	Query Encoding	Candidate Retrieval			Relevance Scoring	Total Time
		Stage 1	Stage 2	Stage 3		
ColBERTv2	16.95	3.65	470.73	13.11	18.90	523.34
TRIAL	18.37	4.12	519.15	13.65	48.68	603.98

Ablation Study

Analysis

- Both token relations and token importance **contribute to improvements** in retrieval accuracy.

	MS MARCO	Arguana	Climate-fever	DBpedia-entity	Fever	Fiqa	HotpotQA	NFCorpus	NQ	Quora	SciDocs	SciFact	TREC-COVID	Touche2022	Avg.
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SPLADE++	-	-	-	-	-	-	-	-	-	-	-	-	-	-	50.70
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TRIAL w/o rel. & importance	45.6	49.6	18.0	48.8	75.9	32.0	62.2	32.3	51.9	80.6	14.0	63.3	77.3	29.6	48.88

Token Relations

Analysis

- Qualitative analysis on BEIR examples shows that **incorporating relation scores improves the ranking** of gold documents.

Query 1: [CLS] [QXQ] who sang the songs in the <i>movie beyond the sea</i> [SEP]		
Document Tokens	Relation Sim.	Gold
[CLS] [DXD] <i>beyond the sea</i> film [SEP] beyond the sea is a 2004 american musical drama film based on the life of singer actor bobby dar ##in [SEP]	↑	O
[CLS] [DXD] <i>beyond the sea</i> film [SEP] the soundtrack album features 18 tracks performed by Kevin space ##y [SEP] starring in the lead role ...	↑	O
[CLS] [DXD] <i>beyond the sea</i> song [SEP] robbie Williams sang a version in the closing credits of finding ne ##mo [SEP]	↓	X
Query 2: [CLS] [QXQ] are <i>local h</i> and <i>for against</i> both from the united states ? [SEP]		
Document Tokens	Relation Sim.	Gold
[CLS] [DXD] local h [SEP] <i>local h</i> is an American rock band originally formed by guitarist and vocalist scott lucas basists matt garcia ...	↑	O
[CLS] [DXD] <i>for against</i> [SEP] <i>for against</i> is a united states post punk dream pop band from lincoln nebraska [SEP] despite numerous ...	↑	O
[CLS] [DXD] local #ea ##ts [SEP] <i>local #ea ##ts</i> is a venture backed restaurant information ... operates in the united states and in 50 ...	↓	X
Query 3: [CLS] [QXQ] were <i>scott derrick ##son</i> and <i>ed wood</i> of the same nationality ? [SEP]		
Document Tokens	Relation Sim.	Gold
[CLS] [DXD] <i>ed wood</i> [SEP] edward davis wood jr October 10 1924 December 10 1978 was an American filmmaker actor writer ...	↑	O
[CLS] [DXD] <i>scott derrick ##son</i> [SEP] scott derrick <i>##son</i> born July 16 1966 is an American director screenwriter and producer [SEP] he lives ...	↑	O
[CLS] [DXD] robert <i>wood</i> Williamson [SEP] robert wood Williamson 1856 12 January 1932 was a British solicitor and anthropologist [SEP]	↓	X

Query 4: [CLS] [QXQ] what is the meaning of <i>x girl friend</i> [SEP]		
Document Tokens	Relation Sim.	Gold
[CLS] [DXD] <i>ex</i> relationship [SEP] in social relationship ... for example one might refer to a music group s ex guitarist or someone s ex <i>friend</i> ...	↑	O
[CLS] [DXD] <i>x ##sca ##pe</i> album [SEP] blue gangs ##ta was written and recorded in ... the remix titled gangs ##ta no <i>friend</i> of mine featured ...	↓	X
Query 5: [CLS] [QXQ] is <i>martin freeman</i> in season 2 of <i>fargo</i> [SEP]		
Document Tokens	Relation Sim.	Gold
[CLS] [DXD] <i>fargo</i> tv series [SEP] the second season in 1979 and starring kirsten dunst Patrick Wilson jesse plemons jean smart and ted danson ...	-	O
[CLS] [DXD] <i>Halloween 4 the return of ...</i> producer paul <i>freeman</i> a friend of akkad with ... under the pseudonym jack <i>martin</i> had written ...	↓	X

Token Importance

Analysis

- Average token weights across different POS tags indicate that TRIAL effectively **learns meaningful token importance** during training.

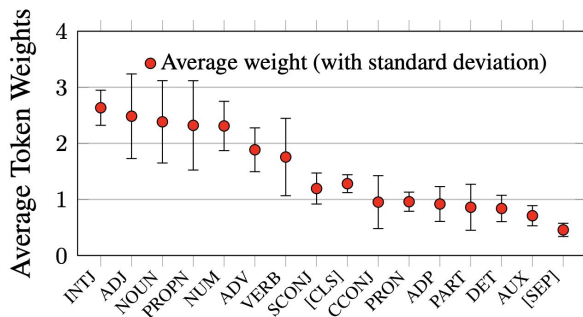


Figure 2: Average token weights for different POS tags in the TRIAL model. Content-rich tokens (e.g., INTJ, ADJ, NOUN) receive higher weights, while functional tokens (e.g., DET, AUX) are assigned lower weights.

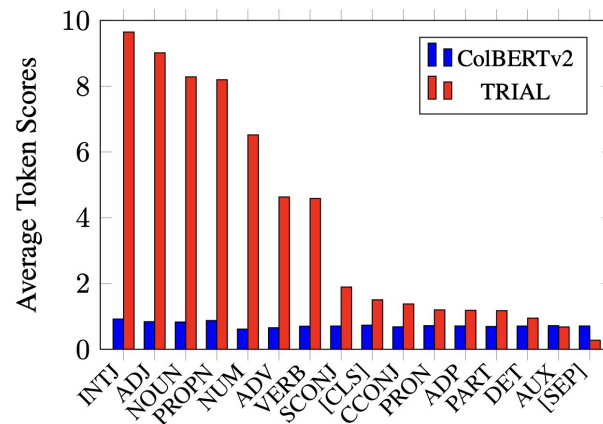


Figure 3: Comparison of scores between ColBERTv2 and TRIAL for different tags, sorted by TRIAL's scores.

Scalability Challenge in Retrieval Speed

Limitation

- While TRIAL **remains practical for real-world applications**, its quadratic complexity with respect to document tokens **may limit scalability for long queries**.

Table 4: Comparison of complexity and estimated FLOPs per query for the scoring functions of ColBERTv2 and TRIAL. FLOPs are computed with $n=32$ (query tokens), $m=300$ (document tokens, and $d=128$ (vector dimension).

	Complexity	Estimated FLOPs/query
$f_{\text{ColBERTv2}}$	nmd	1.22×10^6
f_{TRIAL}	$nmd + nm^2d$	3.69×10^8

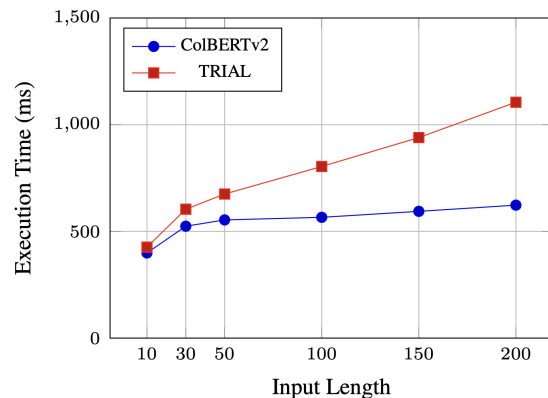


Figure 1: Speed comparison of two models across different input lengths.

Conclusion

- We introduce **TRIAL**, **token relations and importance aware late-interaction**, for accurate text retrieval.
- **TRIAL achieves state-of-the-art retrieval accuracy** across MSMARCO, BEIR, and LoTTE search datasets while maintaining efficiency.
- Analyses show that **modeling token relations and importance jointly enhances** semantic precision and retrieval accuracy.

Thank you